Quantum-Enhanced Resonant Neuro-Modulation for Stage-Gated Dementia Intervention

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Abstract

Cognitive decline in dementia is fundamentally linked to the collapse of hippocampal theta-band network dynamics. Conventional deep brain stimulation (DBS) often utilizes rigid programming, ignoring personalized tissue impedance and neuro-resonant characteristics. Here, we present a resilient framework for quantum-enhanced neuromodulation. Leveraging continued fraction signal decomposition and simulated quantum annealing, we optimize stimulation frequencies to maximize synaptic entrainment. Our results indicate that stage-gated protocols—progressing from initial biomarker audit to high-fidelity resonant plasticity modulation—provide a robust therapeutic pathway for mitigating dementia-induced circuit degradation.

1. Introduction

Dementia is characterized by progressive circuit-level failure within the hippocampal-fornix complex. The Crucian Arch of the fornix acts as a critical bottleneck for memory consolidation; its dysfunction is mirrored in the attenuation of rhythmic neural synchronization. Proactive neuro-modulation via DBS seeks to restore these oscillations by delivering precisely timed electrical pulses.

2. Mathematical Framework

2.1 Axial Magnetic Field Dynamics

The induced stimulation field at a distance z along the vertical axis of a nanoscopically-refined implantable coil is governed by the specialized Biot-Savart derivation:

$$B(z) = (\mu\blacksquare \cdot I \cdot R^2) / (2 \cdot (R^2 + z^2)^{1.5})$$

2.2 Continued Fraction Signal Decomposition

To achieve robust resonance detection amidst biological noise, we decompose the neuro-biosignal $f(\nu)$ using a recursive continued fraction expansion of order n:

$$f(\nu) = a\blacksquare / [b\blacksquare + a\blacksquare / (b\blacksquare + a\blacksquare / (b\blacksquare + \ldots))]$$

2.3 Statistical Congruence Optimization

The optimized therapeutic yield (Y_{opt}) is derived by modulating the base yield Y_{base} with a congruence-based periodic stability factor:

$$Y_{opt} = Y_{base} + ((Y_{base} \cdot 100) \bmod 7.3) / 10$$

3. Methods

Our experimental setup utilizes the NM-QM-Fornix-v2.0 hardware bridge. The protocol transition follows a three-stage optimization path summarized in Table 1.

Stage	Protocol Name	Functional Objective	Target Parameter
I	Biomarker Audit	Baseline Connectivity Check	1.0V (μ=0.1s)
II	Neural Priming	Sub-threshold Sensitization	2.0V (Fixed)
III	Synaptic Entrainment	Resonant Plasticity Mod.	3.5V (Optimal)

Table 1: Stage-Gated DBS Protocol Parameters.

4. Conclusion

By integrating quantum annealing with adaptive resonant feedback, we establish a robust methodology for personalized dementia care. The use of continued fraction decomposition ensures high signal integrity in dynamic neuro-environments.